

Augmented Reality based WayFinder System in Libraries

Aiswarya R

Dept. of Computer Science and
Engineering

Sreepathy Institute of Management and
Technology, Vavanoor, Palakkad
Kerala, India

aiswaryalaxmisaarang@gmail.com

Sreeram V K

Dept. of Computer Science and
Engineering

Sreepathy Institute of Management and
Technology, Vavanoor, Palakkad
Kerala, India

sreeramvk988@gmail.com

Anchana Jinesh K

Dept. of Computer Science and
Engineering

Sreepathy Institute of Management and
Technology, Vavanoor, Palakkad
Kerala, India

anujin02@gmail.com

Sneha VP

Dept. of Computer Science and Engineering

Sreepathy Institute of Management and Technology, Vavanoor,
Palakkad

Kerala, India

snehavpks@gmail.com

Hema P Menon

Dept. of Computer Science and Engineering

Sreepathy Institute of Management and Technology, Vavanoor,
Palakkad

Kerala, India

hemapmenon@simat.ac.in

Abstract—Advancement in technology can be utilized to ease workloads in many scenarios. In libraries, the search for a particular book may take a lot of time, and memorizing the position of each book is not an easy task and may result in reducing a person's interest in visiting the library. Usage of Augmented Reality, helps in visualizing the plots of the library. After combining with GPS, the system will be able to suggest the path to desired book in minimum required time. In libraries, there can be multiple paths to reach a particular book. The proposed system compares the multiple paths to find the best way with minimum distance to reach the desired book. The developed Augmented Reality wayfinding system revolutionizes the library experience by overlaying directional information or real environments through a smartphone camera. Scanning a provided QR code in the library helps the users to access the interface of the system and search for the required book. Innovation of such a system not only simplifies the search for books but also enables customers to check book availability and stock levels via a database. It also suggests alternatives, if the specific book is unavailable. By engaging such on-screen bots overall user experience can be enhanced and workload can be minimized.

Keywords—*Augmented Reality (AR), Global Positioning System (GPS), Navigation, Unity*

INTRODUCTION

Libraries serve as vital hubs of knowledge and information, but navigating their extensive collections can be daunting for visitors, especially newcomers. This abstract introduces an innovative solution – an Augmented Reality (AR) based wayfinder system designed to enhance the library user experience. Leveraging the power of AR technology, this system offers an intuitive and interactive means of guiding patrons within the library space.

By integrating AR features into a user-friendly mobile application, patrons can access real-time, visually enhanced

navigation assistance. The system utilizes markers and sensors strategically placed throughout the library to provide accurate positioning data to the user's device. Users can simply point their smartphones or tablets in different directions to view overlaid digital information, such as directional arrows, floor maps, and highlighted pathways, superimposed on the live camera feed.

This AR-based wayfinder system not only simplifies the process of locating specific books, sections, or amenities within the library but also enriches the overall user experience. It promotes independent exploration, encourages engagement with library resources, and fosters a seamless connection between patrons and the vast reservoir of knowledge housed within the library.

This abstract provides a glimpse into the potential of AR technology in transforming traditional libraries into interactive, digitally augmented spaces, enhancing accessibility, and ensuring that patrons make the most of their library visits.

BACKGROUND STUDY

In the publication by Mi Jeong et al.[1], a hospital was selected as the context for a study on wayfinding in complex environments, and a specially designed instrument was created to comprehend the variables influencing the cognitive process of wayfinding. As discussed in work by Mutis et al.[2] the research explores Augmented Reality (AR) navigation, which requires proficiency in sensor technologies, computer vision, and geographic mapping. Same approach has been used by researchers in their respective works[4][5][9]. Articles highlight features like object detection, real-time tracking, and user interface in their discussions of the application of AR frameworks like ARKit and ARCore.

Using the Wikitude AR SDK and Unity software, a prototype augmented reality application for Geodesy faculty students is described in the work of Cibalic, et.al[3]. Another paper published by Templin, et.al [7] addresses the difficulties in wayfinding in complex environments, especially in an academic setting (University of Technology, Malaysia). By utilising augmented reality technology, spatial planners and information designers hope to improve wayfinding aids and interactions. The article offers information on augmented reality (AR) technology, its hardware elements, and where it falls in relation to virtual reality (VR) and telepresence. A different study by Qiao et.al[10] examines how mobile augmented reality might improve interactions, focusing on striking a balance between virtual data and real-world perspectives to allay security worries. Papers on AR-based navigation systems are numerous. In the Paper by Gu et.al[11] presents a Simultaneous Localization and Mapping (SLAM)-based AR-based indoor navigation system that creates point clouds maps.

To increase the SINS/GPS integrated navigation system's dependability, another paper by Lyu et.al[12] suggests an information fault detection technique. Furthermore, a project of Sayapogu et.al[13] investigating the use of AR for indoor navigation in a large campus makes use of Unity, ARCore, and Navmesh. There is also discussion of intelligent and accurate AR-based navigation systems in the paper published by Wang,[14] with a focus on multifunctional AR navigation features and continuous positioning technology. Finally, paper of AR Navigation Application based on iOS Platform by Wang et.al[15] describes how to create an iOS platform augmented reality navigation application. This application offers real-time GPS-based location, 3D position estimation, and augmented reality scenes superimposed on real-world backgrounds. It consists of modules named Location, Digital Map, Approximate, Estimate, and AR. In conclusion, these efforts together address issues and investigate creative approaches for improved user experience while also helping to comprehend and apply AR technology in a variety of navigational scenarios.

SYSTEM DESIGN

The AR wayfinding system starts with a mobile application installed on the user's smartphone or tablet. The application serves as the user interface through which they interact with the AR wayfinding features. It can be a standalone app or integrated into an existing library app. The mobile device's sensors, such as GPS, accelerometer, gyroscope, and compass, provide essential input to the AR wayfinding system. These sensors help determine the user's location, orientation, and movement, which are crucial for accurate positioning and alignment of the AR content. To enable precise indoor positioning, the AR wayfinding system employs location tracking technologies. These technologies help determine the user's position within the library and track their movements as they navigate.

High-precision GPS modules and inertial sensors are examples of advanced sensor technologies that can be

integrated to greatly improve location tracking accuracy and responsiveness in augmented reality navigation. Furthermore, the integration of computer vision algorithms can improve the system's capacity to identify and comprehend its surroundings, allowing for more accurate localization even in intricate metropolitan settings. This is achieved by utilising real-time environmental data from onboard cameras. Moreover, future positions can be predicted by integrating machine learning algorithms for predictive analysis based on past user motions, which enhances and anticipates the AR navigation experience.

Azimuth, or the horizontal angle of a user's direction with respect to a reference point, is often calculated in augmented reality navigation utilising sensor data from the device. This entails making use of the magnetometers and accelerometers that are integrated within the device. While the accelerometer assists in tilt correction, the magnetometer determines the device's magnetic north direction. Algorithms like sensor fusion or quaternion rotations can compute the azimuth precisely by merging these sensor readings. The computed azimuth is then utilised to ensure that virtual items appropriately match with the user's physical orientation, improving the precision of AR navigation by aligning AR overlays or annotations with the user's real-world view.

Fig.1 gives the overall system architecture. Initially, the developer need to provide a QR code in the library which will help the users to install and use this system. Then the system request to access the current location of the user. After giving permission, we can search for our required book by using the book name. By manipulating user's point of interest, the system will display the AR bot that will guide user to the required book in a better way. This is how the wayfinding is done in this system. The system will acquire data, including the user's present position within the library, once it has gained access to their location. The next step is campus, camera and GPS calibration. It indicates that the sensors for navigation will turn on automatically to perform the wayfinding. The Point of Interest (POI) means the position of the required book. The encoding of POI involves finding the path by using the previously mentioned sensors and the location information of the user and that particular book. Also, the calculations for obtaining the shortest path are carried out. After finding the path, the AR bot will be displayed on the screen to guide the user to reach the book.

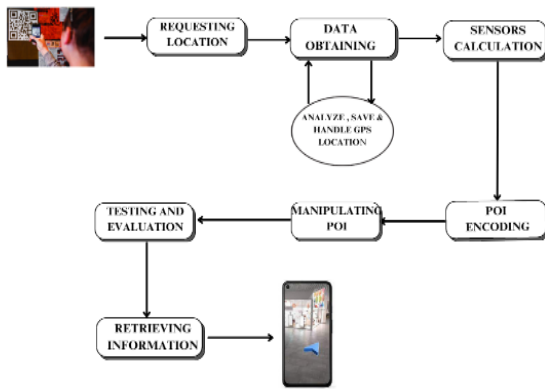


Fig. 1. System Design

Using datasets for Point of Interest (POI) encoding in AR navigation allows for the organised representation of location-specific data, which improves system performance. For a variety of POIs, the dataset contains contextual information, visual descriptors, and geographic coordinates. These datasets usually contain real-world data about businesses, landmarks, and other pertinent locations. Latitude and longitude are two examples of geographic coordinates that are essential for precisely placing points of interest in an augmented reality setting. Using image processing or computer vision techniques, visual descriptors can be obtained that are associated with each point of interest (POI), such as recognisable features or logos.

The dataset is indexed and arranged for speedy retrieval based on the user's location during the encoding process. The AR navigation system can effectively query and display pertinent POIs in the environment thanks to this spatial indexing.

Enabling users to consume augmented reality (AR) content while navigating via AR is essential to a smooth user experience. Through the use of user-friendly interfaces and seamless integration, users can quickly obtain pertinent augmented reality information superimposed on their environment while navigating. In order to provide contextually appropriate data and minimise disruption to the navigation flow, user interfaces must be optimised. Furthermore, integrating voice commands or response gestures might improve user interaction with augmented reality material, facilitating more hands-free and natural information intake. The secret is to seamlessly include augmented reality content into the navigation interface so as to give consumers useful information and direction without sacrificing the navigation's effectiveness.

METHODOLOGY

The design of an effective Augmented Reality navigation system involves a complex interplay of hardware, software, and network components to provide users with seamless and accurate navigation experiences.

A. Hardware Components

- **Smart Devices:** Users interact with the AR system through smartphones, tablets, or AR glasses equipped with cameras, processors, gyroscopes, accelerometers, and GPS modules.
- **Sensors:** Utilize a combination of sensors, including depth sensors, inertial measurement units, and environmental sensors to enhance location tracking and environmental awareness.
- **Inertial Measurement Units (IMUs)** are crucial sensors in AR navigation systems, providing essential information about device orientation and movement, often combining accelerometers, gyroscopes, and magnetometers.

B. Software Components

Unity engine is commonly used for developing augmented reality navigation systems due to its powerful AR development capabilities. Unity supports various AR software development kits (SDKs) such as ARCore for Android and AR Kit for iOS. These SDKs provide essential tools and features for AR development. It allows developers to create and import 3D models of buildings, landmarks, and other navigation elements. These models can be placed in the AR environment, providing users with visual cues and references for navigation. Unity can interface with GPS to determine the user's position on the Earth's surface. The flexible UI system provided by Unity enables developers to create intuitive and user-friendly interfaces for AR navigation. UI elements, such as arrows, waypoints, and textual information, can be dynamically placed and updated in the AR view to guide users effectively. AR navigation systems built with Unity can receive real-time data updates, such as traffic conditions, points of interest, or route changes. With just one RGB camera, Stardust enables incredibly precise displacement, enabling the consumption of AR material at a precise location.

Datasets are essential for training and improving algorithms in AR navigation that enable precise positioning and scene recognition. These datasets usually contain a variety of data, including camera visual data, sensor readings (accelerometer, gyroscope, magnetometer), and GPS locations. Computer vision models are trained with real-world photos and annotated maps to help them identify landmarks and navigate challenging terrain. The creation and testing of AR navigation systems require the availability of extensive and carefully selected datasets in order for them

to produce accurate and dependable location-based augmented reality experiences.

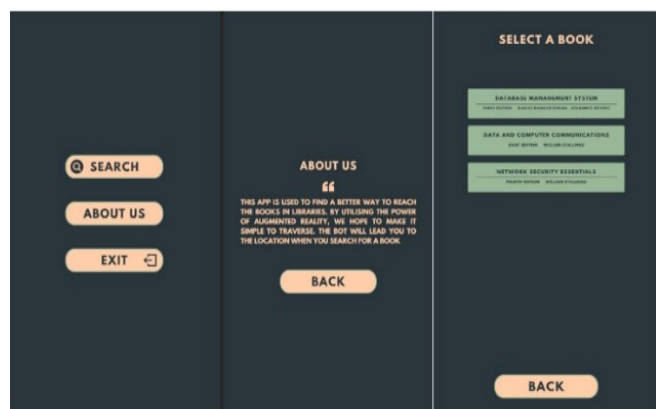


Fig.2.Interface of AR Wayfinder

The interface of the system contains three options such as search, about us and exit. It is pictured in fig.2. Selecting the search button will bring the user to a second interface in which there will be a search bar. User can enter their desired book's name to attain the navigation bot which will guide the user to its position. After obtaining the book, user can exit by simply clicking on the exit button.

In the process of creating an augmented reality (AR) navigation system, the first step involves storing the GPS coordinates of the desired destination in a database. Simultaneously, the system obtains the GPS coordinates of the user's mobile phone, establishing the current location. The next crucial step is to retrieve the stored GPS coordinates of the destination from the database. Using this destination information and the current GPS coordinates, the system calculates the direction, also known as the azimuth, in which the mobile device needs to be oriented for the user to face the desired destination. Once the azimuth is computed, the AR navigation system overlays Points of Interest (POI) onto the user's real-world view, indicating the correct direction in which the mobile device should be pointed to reach the destination. These POI markers are rendered in the user's camera feed, providing a visual guide to the desired location. This interactive and intuitive approach enhances user navigation, making it easier for them to find their way by following the markers aligned with the calculated azimuth.

There are a few important optimisations that can be made to improve the efficiency of an AR navigation system in Unity. First, use level-of-detail techniques to expedite rendering and reduce the amount of resource-intensive graphical effects used. Reduce needless rendering overhead by dynamically rendering only the content that the user can see by utilising Unity's occlusion culling feature. Furthermore, optimise code by putting effective algorithms for processing sensor data into practise. This will guarantee fast and precise location updates without putting system resources at risk. To cut down on load times, use asset bundling and asynchronous loading for your assets. To reduce needless computations, think about putting in place a caching system

for frequently accessed AR data. These optimisations aim to greatly enhance the overall performance and responsiveness of the system by concentrating on effective rendering, astute asset management, and efficient code.

	A	B	C	D	E	F
1	SINO	CODE	BOOK NAME	EDITION	AUTHOR	
2	1	10370	Database Management System	Third	Raghu Ramakrishnan , Johannes Gehrke	
3	2	10356	Data Mining	Third	Jiawei Han, Micheline Kamber, Jian pei	
4	3	11120	Data Mining	Third	Jiawei Han, Micheline Kamber, Jian pei	
5	4	11121	Data Mining	Third	Jiawei Han, Micheline Kamber, Jian pei	
6	5	11122	Data Mining	Third	Jiawei Han, Micheline Kamber, Jian pei	
7	6	11187	Data and Computer Communications	Eight	William Stallings	
8	7	11311	Network Security Essentials	Fourth	William Stallings	
9	8	2056	Artificial Intelligence	Second	Stuart Russel , Peter Norvig	
10	9	11201	System Software	Third	Leland I.Beck	
11	10	10425	Object Software Engineering	Second	Bernad Bruegge, Allen H.Dutoit	
12						
13						
14						

Fig.3. Dataset of Books in Library

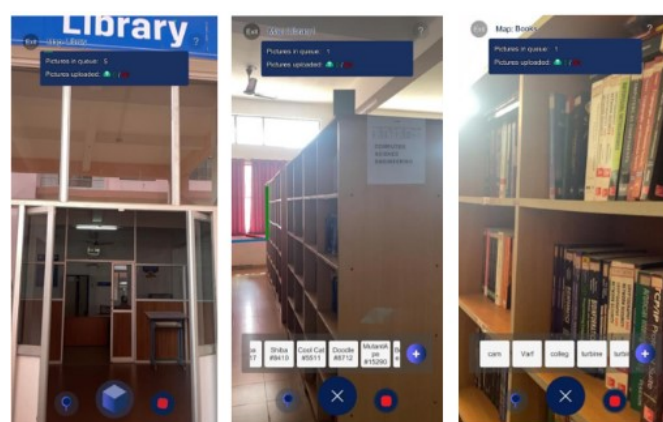


Fig.4.Wayfinding in library

CHALLENGES

There are a number of difficulties in creating an AR wayfinder for a library using datasets and libraries. First of all, because of elements like changing lighting and occlusions, it can be challenging to ensure accurate spatial mapping in a library setting. Integration of external datasets with the unique layout of the library may present calibration challenges. Maintaining the accuracy of the AR annotations is also a challenge in dynamic library spaces with furniture that can be moved around or temporary displays. The integration of an effective dataset management system presents challenges in terms of storage, retrieval, and real-time updates when handling the variety of library content and possibly large datasets.

CONCLUSION AND FUTURE WORKS

AR wayfinding in libraries offers a modern and efficient solution to the challenges of navigating books in large library spaces. It enhances the user experience, improves accessibility, and provides opportunities for interactive engagement with library resources. Users can simply follow the virtual markers or arrows displayed on their devices, eliminating confusion and reducing the time spent searching for items or areas within the library.

The scope of future study in AR wayfinding is broad, encompassing various aspects of user experience, technology development, interaction design, and societal impact. Exploring these areas, researchers can contribute to the advancement of AR wayfinding technologies and shape their practical applications in a wide range of domains, including navigation, tourism.

In future the authors propose to develop the system using an Unreal Engine which would facilitate a more realistic environment. It takes a sophisticated method to create an AR navigation system with Unreal Engine, making use of its powerful visuals and real-time rendering features. This entails utilising AR frameworks such as ARKit or ARCore, incorporating AR functionality into Unreal Engine's Blueprints architecture, and utilising its spatial computing capabilities for real-world mapping and scene interpretation. As a result, the navigating experience is clear and visually appealing.

REFERENCES

- [1] Implementing An Augmented Reality-Enabled Wayfinding System Through Studying User Experience And Requirements in Complex Environments, Mi jeong Kim, Xiangyu Wang, Sooyeon Han, June 2015
- [2] Challenges And Enablers Of Augmented Reality Technology In Situ Walkthrough Applications, Ivan Mutis, Abhijeet Ambekar, January 2020
- [3] Augmented Reality App Exploring And Wayfinding Around Faculty, Cibilic, I, Vukovic, V. and Posloncec-Petric, October 2020
- [4] Effects of Augmented Reality Display Settings on Human Wayfinding Performance, Brian F. Goldiez, Ali M. Ahmad, and Peter A. Hancock , October 2007
- [5] Impact Of Augmented Reality Application On Fashion Industries by Gunasekaran Muthumanickam, Prakash V S , April 2021
- [6] Paper-Based Augmented Reality, Jonathan J. Hull, Berna Erol, Jamey Graham, Qifa Ke, Hidenobu Kishi, Jorge Moraleda, Daniel G. Van Olst, November 2007
- [7] Using Augmented and Virtual Reality (AR/VR) to Support Safe Navigation on Inland and Coastal Water Zones, Tomasz Templin, Dariusz Popielarczyk, Marcin Gryszko, 2022
- [8] Visualising Library Locations with Mobile Augmented Reality (Lib-MAR), Ahmad Nadzri Mohamad, Nurhidayah Hashim, Noor Nazahiah Bakri, Mohd Faizal Hamzah, Mazwani Ayu Mazlan, Tengku Adil Tengku Izhar
- [9] Way-Finding Guidance Using AR Technology, Faris Abduhashish, February 2020
- [10] Web AR: A Promising Future for Mobile Augmented Reality-State of the Art, Challenges and insights, February 2019
- [11] AR-based Navigation Using Hybrid Map, Yanlei Gu; Woranipit Chidsin; Igor Goncharenko
- [12] The Method of Data Fault Detection of the Integrated Navigation System Based on AR Measurement Modling, Xu Lyu
- [13] AR Smart Navigation System-Tina Sayapogu;Kevin Dsa;Priya Kaul 2021
- [14] AR Precision Navigation System Based on Baidu Map API-Zhaoyang Wang; Qirui Tang; Zhou Yang
- [15] AR Navigation Application based on iOS Platform-Xin Wang; Risong Zhang